

Lang Cao

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Education

University of Illinois Urbana-Champaign (UIUC) <i>Doctor of Philosophy in Computer Science</i> - Research Area: Large Language Model, Agent, Reasoning, AI for Health - Advisor: Prof. Yue Guo <i>Master of Science in Computer Science (Thesis)</i>	Urbana, USA Aug. 2022 - May 2028 (Expected) Aug. 2022 - May 2024
Wuhan University of Technology (WUT) <i>Bachelor of Engineering in Software Engineering</i> Rank 1st/79, GPA 3.94/4.0, National Scholarship, Outstanding Graduate	Wuhan, China Sept. 2018 - June 2022

Selected Publications

Google Scholar: <https://scholar.google.com/citations?user=CefzkgpEAAAAJ>

ICLR'26	Lang Cao , Hanbing Liu. “ <i>TableMaster: A Recipe to Advance Table Understanding with Language Models</i> ”. [Paper] [Code]
EMNLP'24 Main	Lang Cao . “ <i>Learn to Refuse: Making Large Language Models More Controllable and Reliable through Knowledge Scope Limitation and Refusal Mechanism</i> ”. [Paper] [Code]
NAACL'24 Main	Lang Cao , Zifeng Wang, Cao Xiao, Jimeng Sun. “ <i>PILOT: Legal Case Outcome Prediction with Case Law</i> ”. [Paper] [Code]
ACL'24 Workshop	Lang Cao . “ <i>GraphReason: Enhancing Reasoning Capabilities of Large Language Models through A Graph-Based Verification Approach</i> ”. [Paper] [Code]
ICLR'26	Pengcheng Jiang, Lang Cao , Ruike Zhu, Minhao Jiang, Yunyi Zhang, Jiaming Shen, Jimeng Sun, and Jiawei Han. “ <i>RAS: Enhanced Knowledge-Intensive LLM Generation with Iterative Retrieval-And-Structure</i> ”. [Paper] [Code]
NeurIPS'24	Pengcheng Jiang, Lang Cao , Cao Xiao, Parminder Bhatia, Jimeng Sun, Jiawei Han. “ <i>KG-FIT: Knowledge Graph Fine-Tuning Upon Open-World Knowledge</i> ”. [Paper] [Code]
COLM'25	Pengcheng Jiang, Jiacheng Lin, Lang Cao , Runchu Tian, SeongKu Kang, Zifeng Wang, Jimeng Sun, Jiawei Han. “ <i>DeepRetrieval: Hacking Real Search Engines and Retrievers with Large Language Models via Reinforcement Learning</i> ”. [Paper] [Code]
ACL'25 Industry	Deyin Yi, Yihao Liu, Lang Cao , Mengyu Zhou, Haoyu Dong, Shi Han, Dongmei Zhang. “ <i>TablePilot: Recommending Human-Preferred Tabular Data Analysis with Large Language Models</i> .” [Paper] [Code]
Nature Communications, 2025	Zifeng Wang, Lang Cao , Qiao Jin, Joey Chan, Nicholas Wan, Behdad Afzali, Hyun-Jin Cho, Chang-In Choi, Mehdi Emamverdi, Manjot K. Gill, Sun-Hyung Kim, Yijia Li, Yi Liu, Yiming Luo, Hanley Ong, Justin F. Rousseau, Irfan Sheikh, Jenny J. Wei, Ziyang Xu, Christopher M. Zallek, Kyungsang Kim, Yifan Peng, Zhiyong Lu, Jimeng Sun. “ <i>A foundation model for human-AI collaboration in medical literature mining</i> ”. [Paper] [Code]
npj Digital Medicine, 2025	Zifeng Wang, Lang Cao , Benjamin Danek, Qiao Jin, Zhiyong Lu, Jimeng Sun. “ <i>Accelerating Clinical Evidence Synthesis with Large Language Models</i> ”. [Paper] [Code]

Preprint, 2026	Lang Cao , Qingyun Chen, Yue Guo. “ <i>EHR-RAG: Bridging Long-Horizon Structured Electronic Health Records and Large Language Models via Enhanced Retrieval-Augmented Generation</i> ”. [Paper] [Code]
ACL’26 Findings	Hanbing Liu*, Lang Cao* , Yang Li. “ <i>RAG or Learning? Understanding the Limits of LLM Adaptation under Continuous Knowledge Drift in the Real World</i> ”. [Paper] [Code]
Preprint, 2025	Lang Cao . “ <i>Chain-of-Alpha: Unleashing the Power of Large Language Models for Alpha Mining in Quantitative Trading</i> ”. [Paper]
Preprint, 2025	Lang Cao , Jingxian Xu, Hanbing Liu, Jinyu Wang, Mengyu Zhou, Haoyu Dong, Shi Han, Dongmei Zhang. “ <i>Formula-R1: Incentivizing LLM Reasoning over Complex Tables with Numerical Computation via Formula-Driven Reinforcement Learning</i> .” [Paper] [Code]
ACL’26 Main	Hanbing Liu, Lang Cao , Yuanyi Ren, Mengyu Zhou, Haoyu Dong, Xiaojun Ma, Shi Han, Dongmei Zhang. “ <i>Not All Tokens Matter: Towards Efficient LLM Reasoning via Token Significance in Reinforcement Learning</i> ”. [Paper] [Code]

Experiences

Amazon <i>Applied Scientist Intern</i> - Research Focus: Agent and Agentic RL - Mentor: Shaoyuan Xu	Seattle, USA May. 2026 – Aug. 2026
University of Illinois Urbana-Champaign <i>Research Assistant</i> Research Focus: Medical Reasoning and AI for Health	Urbana, USA Aug. 2025 – Present
Ubiquant <i>AI and Quant Intern</i> Research Focus: Alpha Mining, AI for Quantitative Trading, LLM Agents of Database and Deep Research.	Shanghai, China May 2025 – Aug. 2025
Microsoft Research <i>Research Intern at Data Knowledge and Intelligence Group</i> - Research Focus: Spreadsheet Intelligence, LLM for Table Understanding, General LLM Reasoning. - Mentor: Haoyu Dong, Mengyu Zhou	Beijing, China Aug. 2024 – June 2025
Tsinghua University <i>Research Assistant at THU-NLP Lab</i> Research Focus: Multi-modal Learning and Reasoning.	Beijing, China Nov. 2024 – Feb. 2025
Keiji AI <i>Research Intern</i> Research Focus: AI for Clinical Trials, Medical AI Agents.	Remote, USA Jan. 2024 – May 2024

Selected Honors & Awards

- Silver Medal, top 5% in Kaggle Common Lit Readability Prize (2021.8)
- Top 2% in Alibaba Tianchi NLP Chinese Pre-training Model Generalization Ability Challenge (2021.1)
- The National Champion of the 2014 FIRST LEGO League Challenge in China (core) (2014.6); Gold Award at the 2016 Asia-Pacific Championship of the FIRST LEGO League Challenge (leader) (2%) (2016.7)

Invited Talks

- “*Thoughts of DeepSeek R1 & Reasoning in Language Models*”, invited by Haoyu Dong at Microsoft Research, (2025.02).
- “*Language Models’ Table Understanding with Formula*”, invited by Mengyu Zhou at Microsoft Research, (2025.04).

Services & Teaching

- Paper Review: ACL Rolling Review (ACL, EMNLP, NAACL), ICLR, ICML, NeurIPS, KDD, AAAI, ICASSP
- Teaching Assistant, Algorithm Design and Analysis, WUT (2020.2-2020.6)

Skills

- Programming: Python (Proficient), C/C++, Java, JavaScript/TypeScript, HTML/CSS, SQL
- Machine Learning Frameworks: PyTorch, TensorFlow, Huggingface Transformers, LangChain, Scikit-learn